Escaping Inequality in India: Role of English-Medium Instruction

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Abstract

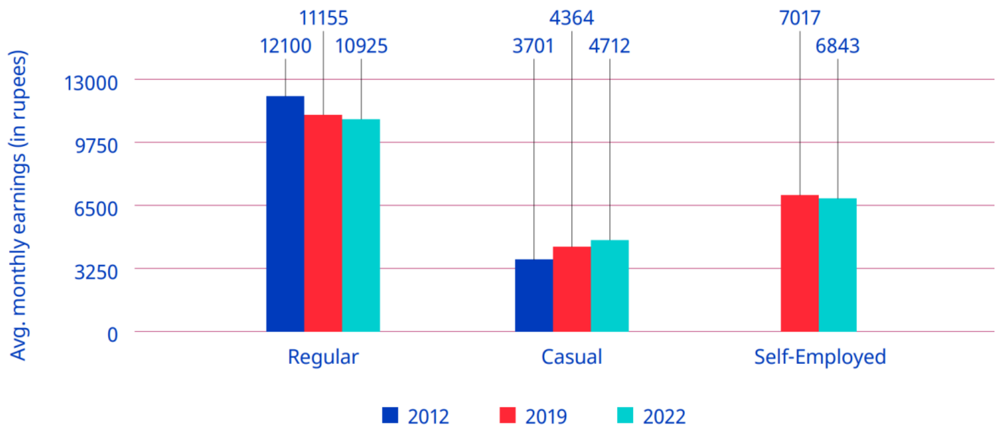
English-medium instruction (EMI), or the teaching of school subjects in English, is often viewed as a potential tool for economic mobility in India, where English skills command substantial wage premia. Yet it remains unclear whether greater exposure to EMI generates broader local development gains. We study whether district-level EMI exposure in 2007, measured as the share of children ages 5-19 who are both enrolled and observed in EMI, affected growth in average household consumption between 2007 and 2018. To address endogeneity, we instrument EMI exposure using a proxy for the opportunity cost of acquiring EMI over Hindi-based schooling: the speaker-weighted mean linguistic distance from Hindi among mapped mother tongues with positive distance in 2001. District-level 2SLS estimates with state-clustered standard errors are positive but insignificant, providing limited evidence that EMI exposure increased local consumption growth over the medium run. This district-level equilibrium analysis is supplemented with an individual-level probit model of selection into education. We conclude by discussing threats to identification and interpretation (namely spatial autocorrelation and migration) and their implications for future work.

1 Introduction and Literature Review

The Indian economy presents a series of paradoxes. The same economy whose GDP has grown at 6% annually on average for decades (International Labour Organization 2024, 2) has also seen stagnation if not losses in wages, employment, and labor force participation (see Figure 1), all while higher education has continued to develop an extremely strong, positive correlation with higher youth *unemployment*.[1] And though the International Labour Organization (ILO) traces this back to a labor supply which lacks the skills needed to fill available job openings (p. 174), high unemployment rates can be found for skilled laborers of all backgrounds, including those who received vocational education and training (Agrawal and Agrawal 2017).

[1] Note that lower levels of education are instead strongly correlated with youth *under*employment (International Labour Organization 2024, 130). There are no winners here.

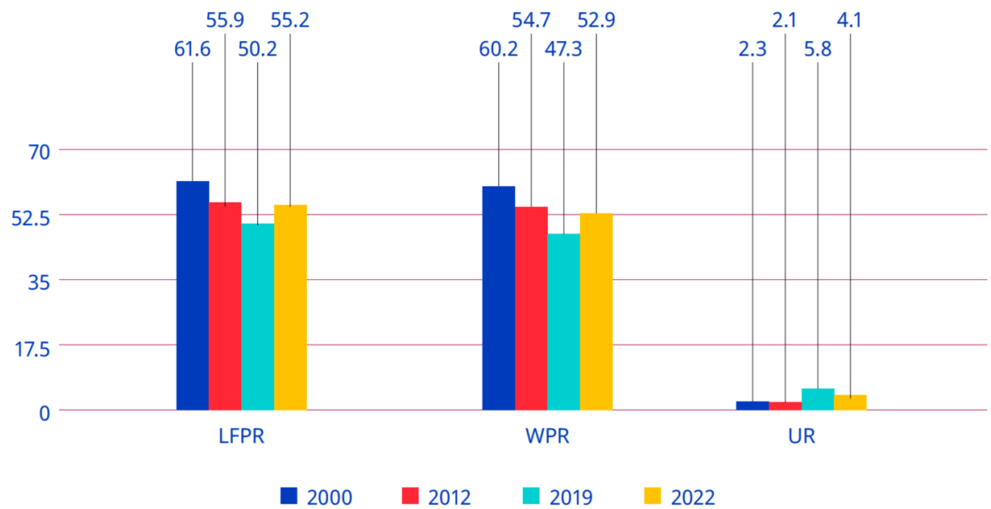
▶ Figure 2.8. Average monthly earnings of regular, casual and self-employed workers, 2012, 2019 and 2022 (rupees, at 2012 prices)



Note: Casual wages also included wages in public works, such as the Mahatma Gandhi National Rural Employment Guarantee. Wages were adjusted using the consumer price index-R and consumer price index-U by taking 2012 as the base year.

Source: Computed from various years of the Employment and Unemployment Survey data and the Periodic Labour Force Survey unit-level data.

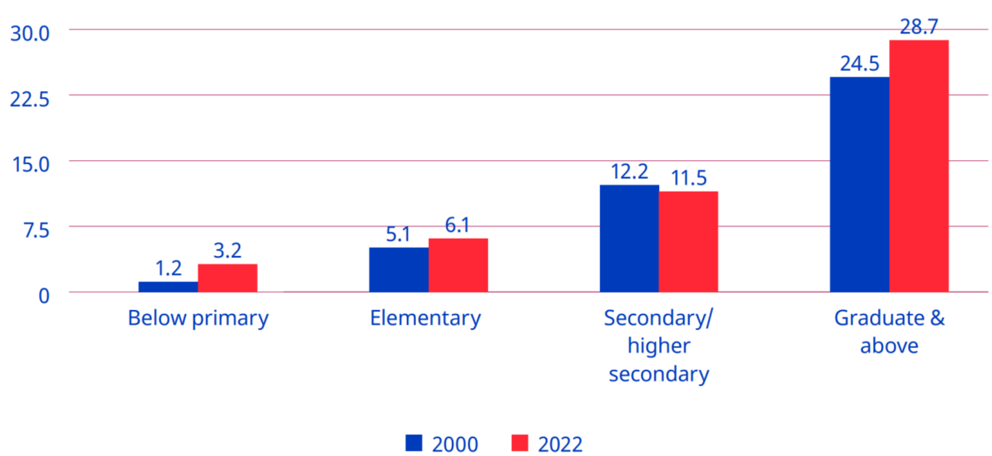
▶ Figure 2.1. Labour force participation rate, worker population ratio and unemployment rate (UPSS) among persons aged 15+ (rural and urban combined), 2000, 2012, 2019 and 2022 (%)



Note: LFPR=labour force participation rate; WPR=worker population ratio; UR=unemployment rate.

Source: Computed from various years of the Employment and Unemployment Survey data and the Periodic Labour Force Survey unit-level data.

▶ Figure 5.14. Unemployment rate for youths, by level of general education (UPSS), 2000 and 2022 (%)



Source: Computed from Employment and Unemployment Survey data for 2000 and Periodic Labour Force Survey data for 2022.

Figure 1: Trends in earnings, labor-force participation, and unemployment (ILO, 2024).

Appreciating this broader view helps reveal the degree to which unemployment, underemployment, and "jobless growth" (Graddol 2010) have, alongside wealth inequality, combined to undermine the degree to which many benefit from India's recent economic growth (Bharti et al. 2024). From here we adopt a bottom-up perspective—what can members of this "many" do to improve their lives in such a context?—and in doing so, our motivating question reveals itself to be two-fold: 1) If I am a household, what can I do to best ensure my children's future success in this context? And 2) if I am a local policy-maker, what can I do to best ensure my constituents' future welfare in this context?

The history of economics highlights *education* as one rich answer to these questions (Mincer 1974; Card 1999; Björklund and Kjellström 2002; Bhandari and Bordoloi 2006). The history of India, however, highlights another: *English*. In its capacity as global lingua franca (Melitz 2016), English is perceived as expanding mobility (Itani et al. 2015) and opening pathways to education at prestigious universities and jobs at wealthy multinational corporations across the world (Ramanathan 2014), particularly in the US (Bedi 2019, 32–33). But combine this with its role as domestic lingua franca (Clingingsmith 2014), and we see the number of intranational pathways English opens up as well: the mass off-shoring of millions of service jobs in the 1990s and the rise of information technology (IT) since have not only driven a rapid rise in India's GDP, but also a flurry of jobs whose higher socioeconomic returns (Chamarbagwala and Sharma 2011; Smokotin et al. 2014) are in large part only accessible to those who speak English (Mankiw and Swagel 2006; Shastry 2012).

Empirical study reveals the combined effect of these mechanisms. Using nationally-representative, household-level panel data, Azam et al. (2013) find that English fluency was, at least up until 2005, associated with a 34% hourly wage premium for men and a 22% premium for women; even for men and women with little English proficiency, the premia were still 13% and 10%. Refeque and Azad (2022) would later replicate this finding and extend it out to 2011-12. The strongest evidence for a causal effect of English skills on wages, however, comes from Chakraborty and Bakshi (2016), for whom West Bengal's 1983-2004 ban on English teaching in public primary schools constituted a natural experiment whose treatment group (those with increased "exposure" to the ban) faced a 26% decrease in future weekly wages relative to the control group (those less "exposed" to the ban)[2].

It is thus by combining our two answers—education and English—that we can begin to understand the demand for *English-medium instruction (EMI)*, or for the teaching of all school subjects in English, currently seen in India (Gooptu and Mukherjee 2023; Lahoti and Mukhopadhyay 2019; Chattopadhay and Roy 2017). "Language skills are human capital," after all (Chiswick and Miller 1995, 4), and "English-language skills [specifically] are a form of human capital" (Azam et al. 2013, 344); if the main mechanism by which education affects wages is human capital, then synthesizing education with English like this seems natural. Perhaps this is why EMI, as both a

[2]We will return to "exposure" again in Section 4.1

tool for social development (National Knowledge Commission 2009) and even as a tool for individual liberation,[3] is so often seen as a source of hope.

But the truth of EMI remains unclear and understudied. On one hand, there are limitations unique to EMI: instruction in one's mother tongue is associated with better academic participation and outcomes, both within India (Mohanty 2022; Nair 2015) and beyond (Bühmann and Trudell 2008), particularly when schooling quality and parents' socioeconomic status are controlled for.[4]

On the other hand, there are limitations unique to the many flaws of the Indian education system: low-effort (Singh 2013), even abusive (Centre for Budget and Policy Studies 2020, 218–22) teachers whose annual absenteeism rates have remained near 25% for decades (Muralidharan et al. 2017; Chaudhury et al. 2006, who also find that only 45% of teachers at any one point in time were engaged in any kind of teaching-related activity)[5] coupled with inefficient resource allocation (Muralidharan and Sundararaman 2013; Gooptu and Mukherjee 2023) and poor commuting infrastructure (Kremer et al. 2005), all alongside schools which actively lie about their medium of instruction (Mathew and Srivastava 2011; Lahoti and Mukhopadhyay 2019), all of which is more commonly found in government schools than private. And yet the mismatch between expectation and reality can be so great that, even among the rural *private* schools surveyed by Lahoti and Mukhopadhyay (2019), "more than half (57%) of the children who [were] supposed to be [in EMI were] actually studying in the dominant regional language" (p. 55). Navigating such an environment requires parents to have enough information and social capital with which to identify higher-quality schooling/tutoring[6]—often the product of being well-educated themselves (Lahoti and Mukhopadhyay 2019; Blimpo et al. 2015)—but also to have a high enough income to afford this higher-quality[7] education (Muralidharan and Kremer 2008; Singh 2013).

The demand for both higher-quality education in general and EMI in particular has, since at least the 2000s, been increasingly met by a supply of 'low-cost'[8] private schools (James and Woodhead 2014; Srivastava 2013;

[3]Particularly from caste and the intergenerationally-rigid distributions of wage, employment, and occupation it enforces, as studied by Madheswaran and Singhari (2016) and Munshi and Rosenzweig (2006). See Mathur (2013) and Mathew and Srivastava (2011) for their advocation of, as well as Vij (2020) for their polemic for, EMI as a means of upwards growth.

[4]The key mechanism here appears to be the effect of exposure to a language at home on a child's ability to learn the language (Nair 2015; Erling et al. 2016). The dissimilarity between the languages spoken at home and the colonial languages chosen as lingua franca and media of instruction across the extremely linguistically diverse sub-Saharan Africa helps explain, per Laitin and Ramachandran (2022), the low rates of absolute learning and thus low rates of development in the region. Another potential mechanism, once hotly debated, is that childhood bilingualism generally confuses children, and that they most effectively learn with only one language while young (Hakuta 1986). The true effects of such bilingualism, however, seem to be much more nuanced; see Bialystok et al. (2012) for a review and for the effects of bilingualism into adulthood and old age.

[5]The long-term consistency of this trend reflects the longevity of the civil service protections which produce it: limits on teacher hiring/firing power in government schools enable qualified teachers to extract such rents (Chaudhury et al. 2006). Even when rarely-absent teachers rationally advocate for such protections, they end up further deepening the culture of absence which Basu (2006) finds to explain the significant state-by-state heterogeneity in teacher absenteeism rates. The simple act of increasing the frequency of teacher monitoring, however, is enough to drastically reduce absenteeism (Kremer et al. 2005; Muralidharan et al. 2017).

[6]The best evidence for this comes from Andrabi et al. (2017) and Afridi et al. (2020), whose experimental treatments of education information provisioned to parents led to a more efficient market and improved student outcomes, particularly in the private schools nudged to improve quality by the treatment.

[7]Which in practice often equals private schooling; see Tooley et al. (2011).

[8]'Low cost' may be a misnomer: poor households in the Delhi National Capital who sent their children to private school did so at the expense of 7-20% of their monthly incomes (Endow 2019).

Ramanathan 2016; Chattopadhay and Roy 2017).[9] Such schools lack the reputation typically afforded to higher-end private schools, meaning their students lack the social capital of other private school students (Ramanathan 2016). Conversely, more disadvantaged students are priced out of all schools except the free (if not negatively priced, see Muralidharan 2019) government schools. Note that these schools are not only associated with a lower quality of education (Muralidharan and Kremer 2008; Srivastava 2013; Muralidharan and Sundararaman 2015; Raju 2023), they are also associated with the most obvious signals of education quality, visible even to low-information parents, being worse-off e.g., worse facilities (Srivastava 2013; Central Square Foundation 2020, 54). Here we have evidence that school search ends with an inequality-amplifying positive assortative matching (Muralidharan 2019). Yet even amongst this new 'middle class' of schools, issues with EMI persist: the survey of Lahoti and Mukhopadhyay (2019) and the ethnographic study of Bhattacharya (2013) find children in low-cost private schools failing to learn both English as well as the subjects taught in it, further corroboration of the findings of Bühmann and Trudell (2008), Nair (2015), and Mohanty (2022). The reasons for this are likely two-fold: compared to parents who enroll their children in high-end private schools, those who select into low-cost private schools are typically 1) lacking in information and social capital, thus "[making] misguided evaluations of school quality based on visible factors" (Muralidharan and Sundararaman 2015, 1061–62); and 2) do not provide their children with as much community and out-of-classroom support for English (particularly when compared to "elite families" who often use English as a second language even within the family (Gupta 1997, 54–55)), preventing their oral skills from developing to the point where they can engage in class beyond rote memorization of the teacher's answers (Bhattacharya 2013; Treffers-Daller et al. 2022), to the extent such engagement is allowed (Brinkmann 2015).[10]

Given all these mediators, nuances, and negations of its perceived effect, can EMI truly be seen as a source of hope? Should households turn to it to invest in their children's future? Does greater enrollment in EMI aid future development? Should policy-makers be promoting it as a tool for local development? To gain insight into these questions, we seek to answer the following one:

> What is the effect of exposure to English-medium instruction enrollment in 2007-08 on the growth of local consumption from 2007-08 to 2017-18?

First, note the improvements made here over the previous literature. The setting of 2007-2018 may be more

[9] The demand for higher quality education almost always takes precedence over the demand for EMI, however. In 2014, for example, Joshi and Kumar (2017) show that 76% of households who chose private education did so for either a "better environment of learning" or to get a better education than a government school. Only 15.4%, however, listed EMI. Still, aside from private schooling, EMI seems to be the most frequently used proxy for high quality education

[10] It's worth noting our exclusion of the results from Muralidharan and Sundararaman (2015), despite it being a particularly noteworthy paper on the limitations of private schooling. By studying a randomized controlled trial of a private EMI voucher experiment, they discover that private schools did *not* lead to improvements in educational outcomes (although they did significantly lower cost). Tooley (2016), however, points out that this result may be driven by the fact that instructions for non-language subject exams were given in Telugu for government schools and English for private schools. He then finds "suggestive" evidence of what results would look like if all children took the same test (p. 580), and from there demonstrates a large, statistically significant advantage for private education.

relevant today than the settings of similar literature, all of which are closer to the advent of service job offshoring in the 1990s (e.g., Refeque and Azad 2022). Additionally, note that our response variable here is consumption, not wages. India holds a vast number of informal and self-employed individuals, meaning its labor force varies wildly in the degree to which wages reflect material well-being.[11]

Second, we must highlight that unlike much of this literature, we lack household panel data. We will thus follow previous papers (e.g., Martin et al. 2017) and aggregate data at the district level.[12] More specifically, we will take cross-sectional data from 2007-08 (National Sample Survey Office 2011) and 2017-18 (National Sample Survey Office 2022), aggregate each at the district level separately, and match districts across the time periods to construct a district pseudo-panel. District-level variables are developed in Section 4.1 and the district matching method is explained in Section A.2 .

Our empirical methodology will then proceed in two parts. In Section 3, we study the composition of our key "EMI exposure" measure by estimating an individual-level probit of school participation in 2007-08. In Section 4, we move to the district level and implement two-stage least squares (2SLS): the weighted average of linguistic distance from Hindi in 2001 will be used to instrument for EMI exposure, onto whose fitted values we regress the change in log person-weighted real consumption.[13] The choice of instrument is discussed in Section 4.2.

Our 2SLS results will, at best, reflect the medium-run local general equilibrium effects of EMI exposure. On one hand, by aggregating away all individual or household-level variation, the ecological fallacy renders our 2SLS estimates uninterpretable at these lower levels. Additionally, school-going children in 2007-08 will still be quite young in 2017-18, and many may still be in school (particularly in more developed districts).[14]

In the preferred specification, the estimated effect of EMI exposure on person-weighted real consumption growth is statistically imprecise (Section 4.4), in line with previous literature on EMI and Indian education quality. Note, however, that our outcome includes households not directly exposed to EMI, meaning the effect we're studying would likely be inherently small regardless. Migration and spatial spillovers may also attenuate our results; evidence which both challenges and supports this is discussed in detail in Section A.2.2).

[11]See Shrivastav and Husain (2026) and Bahl and Sharma (2024) for the frequency and wage effects of informality respectively. Wages may still be more responsive to EMI in the medium-run window we study, however. See Section A.1 for more.

[12]A district in India is akin to a county in the United States, but with an average population of 2 million. Table 4 contains more information on district populations.

[13]Future work may incorporate both education selection and EMI exposure as endogenous regressors in a first-difference (FD) 2SLS design. The district-level aggregates of Table 1 may be able to function as education supply-shifting instruments.

[14]Cohort-specific measures of EMI exposure may thus be incorporated into future drafts, as may robustness checks that use school-reported data on medium of instruction to construct EMI exposure—with the aforementioned caveat that schools may lie about having EMI (Mathew and Srivastava 2011; Lahoti and Mukhopadhyay 2019). Such surveys may also reveal whether EMI exposure is rigid over time. If so, this could allow us to expand the window of time over which we can interpret our results. Rigidity could also rule out the agglomeration of EMI discussed in Section 4.4 as a potential attenuating force behind our results.

2 Data Sources

The sole 2001 variable, linguistic distance from Hindi, comes from the 2001 Census of India (Office of the Registrar General and Census Commissioner of India 2001).

Geospatial data used to construct the maps in this paper as well as all spatial autocorrelation measures is sourced from Bhatia (2020), which is itself an adaptation of Meyers (2020). Our methods for tracking districts across time (see Section A.2.1) begin with data from India State Stories (2025) and Jaacks et al. (2020).

All variables reflecting 2017-18 data come from the 75th Round of the National Sample Survey (NSS), titled "Household Social Consumption: Education" (National Sample Survey Office 2022). All 2007-08 variables but one come from the 64th round of the NSS, on "Participation and Expenditure in Education" (National Sample Survey Office 2011); the sole exception to this is the percentage of homes which are pucca (i.e., permanent), which is instead sourced from a separate schedule of the 64th NSS, "Household Consumer Expenditure" (National Statistical Office 2011). Response rates in all of the surveys were above 96%.

Sample weights were applied wherever applicable. All district-level aggregates were formed as sample-weighted averages. The education participation probit of Section 3 used survey-weighted generalized linear models and design-based standard errors.[15]. Summary statistics for variables used in the probit are given in Table 1 and Table 2, whereas Table 4 contains summary statistics for variables used in the 2SLS estimation.

Except for certain individual-level variables in the education probit, we will be aggregating variables at the district level. As will be shown in Table 4, the average population of a district in either sample period (2007-08 and 2017-18) is 2 million. In the survey samples, however, most districts have between 400 and 1200 people represented in the data, with the exact number varying in line with a district's population.

2.1 Data Issues: Representativeness and Quality

Despite the complicated methodology and rigor underlying each NSS round (see, e.g., Mahalanobis 1954), three key issues with NSS data still exist. We address them in increasing order of severity.

The first issue is the observation from Deolalikar and Dubey (2008) that "the NSS has sometimes changed its data collection methodology midstream, and this has affected the comparability of NSS estimates over time" (p. 410). The examples they and the Ministry of Statistics and Program Implementation (2024) mention predate all of our data, whereas the change described by the Ministry of Statistics and Program Implementation (2024) (that the base year for calculating national accounts has shifted from 2004-05 to 2011-12) should be mostly irrelevant

[15] Clusters were formed at the primary sampling unit, which were rural villages and urban blocks. These clusters were nested, or defined within, substrata which were themselves nested within strata. (Strata generally corresponded to districts, whereas substrata generally corresponded to each side of the rural/urban partition of the district. Multiple urban substrata were allowed for particularly dense areas, with the largest number of substrata within a district being 29.) All were accounted for in the probit.

for us. More relevant, however, is the politicization of the NSS documented by Bhattacharya (2023), which may affect our 2017-18 measures: Bhattacharya begins this period of NSS history by describing a major revision to GDP figures in 2014-15 which was generally misaligned with most other economic indicators. Making this even more suspicious, however, was the state's unwillingness to reveal any information on the reasons or methodology behind this change. Revelations of ghost firms in the NSS database used to calculate the new GDP series, plus a leak of poor unemployment data in 2017-18 which was not officially released until after the Lok Sabha election in 2019, hurt the trustworthiness of our data; Subramanian (2019) ultimately finds that, driven by an improper use of value-added vs. volume metrics, informal activity proxies, and deflators within industries, the NSS Office had overestimated GDP growth rates by 2.5 percentage points.

More worrying is the observation by Lucas (2021b) that "the NSS is not representative at the district level" (p. 329). While Lucas does not provide a citation for this, it still may be true. Samrat (n.d.) writes about a conference on Oct. 9th, 2025 where the National Statistics Office had touted its ability to, among other things, provide district-level estimates; the language is unclear on whether this means district-level estimates will now be provided in NSS reports or that district-level estimates can now be meaningfully calculated from NSS data. Documentation from the National Sample Survey Organisation (2001) states that NSS sampling is built "in order to ensure proportionate representation or pre-fixed numbers of sample households from each group / stratum," which we henceforth use synonymously (p. 15). Defining proportionate representation at the stratum level has been an NSS tradition since Mahalanobis (1954, 274–76), but the benefit of the NSS rounds that we use is that strata are now defined within sectors, or the rural and urban areas of each district (with multiple urban sectors if population density is high).

Consider National Sample Survey Office (2011), the source of most of our 2007-08 measures. Its sampling begins at the national level, where the total number of first-stage units (FSUs) is set, each FSU corresponding to a village or Panchayat ward in rural areas, or a block or a town/outgrowth (OG) in urban areas. These FSUs are then allocated in proportion to population in the following order: first, between the rural vs. urban sectors of the state, with a slight weight to urban; then to strata; then sub-stratum; and from there, FSUs are selected. In rural areas, selection was done using Probability Proportional to Size With Replacement (PPSWR), with the size being the population in the 2001 census. Simple Random Sampling Without Replacement (SRSWOR) was used for urban areas where FSUs were blocks, and PPSWR (with weights from the 2001 census) was used where FSUs were towns/OGs. The documentation for National Sample Survey Office (2022), the source of our 2017-18 measures, makes it clear that FSUs are allocated first to state, then to within-state region (a more recent addition), then to districts, strata, sub-strata, sub-rounds, sub-samples, and FOD sub-regions.

Even then, if each district gets rural and urban sectors, and if sample FSUs are selected from within each, it's possible to make the claim that district-level estimates are representative; if samples within strict subsets of the district are representative, why should districts not be able to inherit that?

Where representativeness at the district-level could fall apart, however, is in FSU selection. Four FSUs are selected from each sub-stratum, so if a district has only one sub-stratum, then only four FSUs may be selected in that district. Additionally, the use of PPSWR (with replacement) in rural strata or for town/OG FSUs means that the same FSU might theoretically be selected more than once (Brus 2023). The design may thus be sufficient for producing reliable state-level estimates, and perhaps even for major districts; but it may be weak for smaller districts with low sample sizes.

Finally we have the most difficult issue to address: general data quality issues. The response rate to NSS surveys systematically decreases in a household's wealth and income, helping ensure endogeneity and bias behind what was reported and what was not (Ravi 2023). For surveys which use the 2011 census as their sampling frame (which includes our 2017-18 data), meanwhile, all of the rural, working age (15-29), and SC (Scheduled Caste) populations were found to have been overestimated significantly. Contrasting 2011-12 econometric values with nation-wide shares from the 2011 Census, Ravi et al. (2023) find that a sample size of 464,960 becomes a bias-adjusted sample size[16] of 324, 240, or 536 given rural, SC, and working age overestimation respectively. This is a 99.9% reduction in statistical efficiency across the board (Ravi et al. 2023, 11).

3 The Composition of Education Participation

3.1 Model and Variables

We are interested in how consumption growth is affected by EMI exposure (EMIE), defined over the full population of children ages 5-19:

$$\text{EMIE}_d = 100 \times \frac{\text{Num. children enrolled in EMI}}{\text{Num. children ages 5–19}} \text{ in district } d.$$

This all-child exposure combines the extensive enrollment margin with the choice of English-medium instruction conditional on enrollment. When medium is observed for every enrolled child, it equals the enrollment rate multiplied by the EMI share among enrolled children. To understand the first margin directly, we estimate the

[16] "Intuitively, the bias-adjusted sample size is the size of the simple random sample that would have generated the same mean squared error" (Ravi et al. 2023, 3)

following probit model for whether child i is enrolled:

$$\begin{aligned}\Pr(\text{Enrolled}_i = 1 \mid X_i,\, Z_{d(i)}) = \Phi\Big(&\beta_0 + \beta_1 \,\text{Age}_i + \beta_2 \,\text{Female}_i + \beta_3 \,\text{HHSize}_i + \beta_4 \,\text{Urban}_i \\ &+ \sum_{r\in R} \gamma_r \,\text{Religion}_{ir} + \sum_{s\in S} \delta_s \,\text{SocialGroup}_{is} + \sum_{d\in D} \theta_d \,\text{SchoolDist}_{id} \\ &+ \sum_{f\in F} \psi_f, \text{FatherEduc}_{if} + \sum_{u\in U} \eta_u \,\overline{u}_{d(i)} + \phi \,\overline{\text{EnrollmentCost}}_{d(i)}\Big) \end{aligned} \tag{1}$$

where

$$\begin{aligned} R &= \{\text{Muslim, Christian, Sikh, Jain, Buddhist, Zoroastrian, None}\}, \\ S &= \{\text{Scheduled Tribe, Scheduled Caste, OBC}\}, \\ D &= \{\text{1–2km, 2–3km, 3–5km}, \geq \text{5km}\}, \\ F &= \{\text{Illiterate; Literate, no school; Literate, school} < \text{primary;} \\ &\quad\ \text{Primary, Upper primary, Secondary, Higher secondary, Postsecondary+}\}, \\ U &= \{\text{Educ. free available, Tuition waived, Scholarship/Stipend,} \\ &\quad\ \text{Textbooks received, Stationery received, Mid-day meal, etc.}\} \end{aligned}$$

and

$$\begin{aligned} \overline{u}_{d(i)} &= \text{Average value of } u \in U \text{ in district } d \text{ amongst enrolled children,} \\ \overline{\text{EnrollmentCost}}_{d(i)} &= \text{Average value of EnrollmentCost in } d \text{ amongst enrolled children.} \end{aligned}$$

There are two kinds of regressors here: those in X_i, the variables unique to child i, and $Z_{d(i)} = \{\overline{u}_{d(i)} : u \in U\} \cup \{\overline{\text{EnrollmentCost}}_{d(i)}\}$, the variables which have been aggregated at the level of the district $d(i)$ where child i lives. Without this aggregation, neither u nor EnrollmentCost, both endogenous to enrollment, would be well-defined for children not in school. Our aggregation can thus be understood as a way of handling missingness; whereas EnrollmentCost_i was missing for 120,062 children (as reflected in Table 1), $\overline{\text{EnrollmentCost}}_{d(i)}$ was missing for 7,109. Observations still missing a relevant variable were then dropped.[17]

The district-level averages can be interpreted as reflections of local supply-side factors, namely the general availability or generosity of educational inputs in a child's local environment. Their marginal effects (see Table 3) may

[17] Three such variables had missing values before dropping: the district-average $\overline{\text{EnrollmentCost}}_{d(i)}$ (7109 missing values), the distance from the nearest primary class (312), and the father's education proxy (67). We find that none of these variables are MCAR (missing completely at random) under logistic regressions with missingness indiciators and adjusted p-values (per Benjamini and Hochberg 1995). The lowest McFadden pseudo R^2 among the three was a 0.246 on father's education, which had three predictors significant at a level of 0.05 (that is, three variables in the data could predict its missingness with adjusted p-values below the 0.05 threshold). The highest pseudo R^2 was a 0.31 for $\overline{\text{EnrollmentCost}}_{d(i)}$, which had nine significant predictors. Note that prior to this, many other NAs in $\overline{\text{EnrollmentCost}}_{d(i)}$ were handled using codebook-informed aggregations and redefinitions of the variables summed to derive EnrollmentCost_i.

reflect how changes in these district-level characteristics affect the probability a child enrolls in education—or in other words, the elasticity of enrollment with respect to local schooling conditions.[18]

Following Azam et al. (2013), Munshi and Rosenzweig (2006), and Singh (2013), we use father's education to proxy a child's ability before schooling. We use the father despite evidence indicating that the mother's education is more strongly associated with early life human capital development.[19] Taking the average of the mother's and father's education is not possible as education level is recorded by the National Sample Survey Office (2011) as a categorical variable. Future work may attempt to create an index variable for this.

The National Sample Survey Office (2011) do not explicitly identify parent-child relationships. Instead, they only provide each person's relationship to the head of their household. Father's education is thus proxied by the education of the head of the household if they are male (average age here is 14); else as the married child of the head if they are male (married men often live with their elderly parents; average age of 18.1); else as the parent/parent-in-law of the head if they are male (average age of 13.5). We do not include male spouses of the head, whose average age is 11.8 and share of children (aged <18) is 0.896 in our data.[20]

Summary statistics for numeric variables are given in Table 1, and for categorical variables in Table 2.

[18]Future work will likely use these district-level averages as instruments for enrollment, an endogenous variable to be included alongside EMIE in a first-difference (FD) 2SLS framework. It may be possible to use this probit as a control function or in a two-stage residual inclusion (2SRI) estimation.

[19]Card (1999) shows that in the US, among people born from before 1920 to 1964, mother's education better predicts school completion among Black people, whereas father's education better predicts for cohorts born 1955-1964. In general, however, he finds mother's education to be the the slightly better predictor of school completion. It's interesting to note that this is a quantitative outcome: one might also expect mother's education to better predict the *quality* of early childhood investment too, with father's education relating more to the quantity of early childhood investment.

[20]This is a clear indication of severe, systematic misclassification error. Further checks on the presence of misclassification imply that the codes for unmarried children and spouses of heads were simply flipped; the average age of male "unmarried children," for example, is 18.1 in the data. Future work will either incorporate the mislabeled "unmarried children" into this proxy chain for father's education, or use the household head's education alone as proxy.

Table 1: Summary Statistics for Enrollment Participation Model (Numeric Variables)

Variable	Min	1Q	Med	3Q	Max	Mean	SD	N
Age	5.00	8.00	12.00	16.00	19.00	11.89	4.22	127181
Household size	1.00	4.00	5.00	7.00	30.00	5.91	2.40	127181
Enrollment cost (Rs.)	0.00	2,315.00	4,900.00	8,750.00	338,300.00	7,050.07	9,285.69	7181
District-level aggregates:								
Educ. free available? (Yes = 1)	0.00	0.48	0.69	0.85	1.00	0.65	0.23	127181
Tuition waived?	0.00	0.00	0.01	0.02	0.48	0.02	0.04	127181
Scholarship/Stipend?	0.00	0.01	0.05	0.20	0.90	0.13	0.17	127181
Textbooks received?	0.00	0.37	0.52	0.66	0.97	0.51	0.20	127181
Stationery received?	0.00	0.01	0.04	0.11	0.95	0.08	0.12	127181
Mid-day meal or more received?	0.02	0.33	0.42	0.55	0.90	0.43	0.17	127181
Avg. district enrollment cost	84.55	3060.61	5158.42	7661.91	58055.64	6010.16	4352.63	120072

Note: Min. = minimum; 1Q = first quartile; Med. = median; 3Q = third quartile; Max. = maximum; Mean = arithmetic mean; SD = standard deviation; N = number of observations.

Table 2: Summary Statistics for Enrollment Participation Model
(Categorical Variables)

Variable	Values	Mode	Pct. Mode	Least Freq.	Pct. Least Freq.	N
Sex	Male, Female	Female	53.4	Male	46.6	127181
Religion	Hindu, Muslim, Christian, Sikh, Jain, Buddhist, Zoroastrian, Other	Hindu	75.2	Other	0	127181
Social group	Other, Scheduled Tribe, Scheduled Caste, Other Backward Class	Other Backward Class	39.5	Scheduled Tribe	14.2	127181
Urban	Rural, Urban	Rural	67.5	Urban	32.5	127181
Distance of nearest primary class	d<1km, 1km <= d <2kms, 2kms<= d <3kms, 3kms <= d <5kms, d>=5kms	d<1km	91.7	3kms <= d <5kms	0.1	126869
Father's education	Illiterate, Literate, no school, Literate, school < primary, Primary, Upper primary, Secondary, Higher secondary, Postsecondary+	Illiterate	55.8	Literate, no school	1.4	121978

Note: Values = all possible values; Mode = most frequent value; Pct. Mode = percent of observations taking the modal value; Least Freq. = least frequent value; Pct. Least Freq. = percent of observations taking the least frequent value; N = number of observations.

3.2 Results and Discussion

Average marginal effects for numeric variables and counterfactual comparisons (relative to the reference level) for categorical variables (both referred to as AME henceforth) are reported in Table 3, which has been derived using the work of Arel-Bundock et al. (2024) In line with advice from Nowosad and Tennekes (2021), we report the AME instead of marginal effects at the mean.

Table 3 has been calculated over all observations, with standard errors clustered at the rural village and urban block level (no district clustering) derived using the delta method and automatic differentiation.[21] All reported values are probabilities: that AME_{Age} = -0.028 in the table below thus means that, on average, a one-year increase in age is associated with a 2.834 percentage point decrease in the probability of being enrolled by, holding all else constant (Hansen 2022). The counterfactual comparison $\text{AME}_{\text{Muslim}}$ = -0.126, meanwhile, where 'Hindu' is the reference category to 'Muslim', means that the average change in the probability of a Hindu child being enrolled if they were hypothetically made Muslim is -0.126, keeping all else fixed at observed values.[22]

Table 3: Average Marginal Effects and Counterfactual Comparisons for Enrollment Probit

	Enrolled (1 = yes)
	(1)
Age (years)	-0.028***
	(0.000)
Female (ref: Male)	0.044***
	(0.003)
Household size	-0.004***
	(0.001)
Religion: Muslim (ref: Hindu)	-0.126***
	(0.007)
Religion: Christian	-0.043**
	(0.014)
Religion: Sikh	0.040
	(0.025)
Religion: Jain	-0.001
	(0.012)
Religion: Buddhist	0.047
	(0.024)
Religion: Zoroastrian	0.101**

[21] Dawood and Megahed (2023) provide a good treatment of automatic differentiation, describing it as a process which is "neither numeric nor symbolic, nor is it a combination of both.... In contrast to the more traditional numerical methods based on finite differences, auto-differentiation is 'in theory' exact, and in comparison to symbolic algorithms, it is computationally inexpensive." The `marginaleffects` package itself claims automatic differentiation is both faster and more accurate for some models (Arel-Bundock et al. 2024).

[22] Such interpretations are possible because AME is averaged across all observations, making them more generalizable than metrics like marginal effects at the mean (Nguyen 2020). But their interpretability is still also dependent on a linearity assumption which may not be valid. Future work may thus abide by the argument of Scholbeck et al. (2024), that no one single metric can be used to summarize a non-linear prediction function's feature effects, and may instead report conditional feature effect estimates by applying their concept of forward marginal effects within population subgroups.

Table 3: Average Marginal Effects and Counterfactual Comparisons for Enrollment Probit *(continued)*

	(1)
	(0.034)
Religion: Other	0.252***
	(0.003)
Social group: Scheduled Tribe (ref: Other)	-0.102***
	(0.008)
Social group: Scheduled Caste	-0.075***
	(0.006)
Social group: Other Backward Class	-0.029***
	(0.005)
Urban (ref: Rural)	-0.003
	(0.005)
Distance 1–2km (ref: $<$1km)	-0.077**
	(0.026)
Distance 2–3km	-0.001
	(0.010)
Distance 3–5km	0.004
	(0.066)
Distance $>$ 5km	0.036
	(0.035)
Father's educ.: Literate, no school (ref: Illiterate)	0.097***
	(0.017)
Father's educ.: Literate, school $<$ primary	0.127***
	(0.006)
Father's educ.: Primary	0.142***
	(0.005)
Father's educ.: Upper primary	0.195***
	(0.006)
Father's educ.: Secondary	0.212***
	(0.007)
Father's educ.: Higher secondary	0.247***
	(0.008)
Father's educ.: Postsecondary+	0.245***
	(0.009)
Educ. free available (ref: No)	-0.089***
	(0.015)
Tuition waiver received	-0.083
	(0.057)
Scholarship/Stipend received	0.030*
	(0.012)
Textbook(s) received	0.067***
	(0.018)
Stationery received	0.013
	(0.020)
Mid-day meal, etc. received	-0.011
	(0.019)
Enrollment cost (Rs.)	0.000
	(0.000)
Observations	127181

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
NSS 64th round; design-based SEs in parentheses.

The goal of this paper is to study how well household optimization (through education participation and EMI enrollment) and local policy changes (aimed at shifting education supply or EMI enrollment) can lead to upward mobility given the context they exist in: one of inequality and jobless growth. In our results we do indeed find supply and social structure to be meaningful predictors of how households optimize via education enrollment.

We first look at household-level determinants of who gets to stay in school. Our data is of children aged 5-19, so as a 5-18 year old child grows one year older, the probability they stay enrolled in school decreases by -2.834 percentage points on average, indicative of either economic necessity, the opportunity cost of school, or even disillusionment with school quality growing in age.[23] We also see that each additional household member lowers enrollment probability by -0.368 percent. A negative value is expected—household resources should dilute as size increases—but the slight value and high significance are notable given the quantity-quality tradeoff of e.g., Becker (1969), plus empirical tests of it in India. Kumar and Kugler (2011) find that, by using children's educational attainment as a proxy for quality, cross-sectional dataset from 2007-08 indicates the tradeoff is heterogeneous in India, stronger given rural areas, low-caste children, illiterate mothers, and poor children. Conversely, by instrumenting family size Onur and Velamuri (2021) remove the quantity-quality tradeoff entirely from their panel data, whether quality is proxied by educational expenditures on a child or test scores. Enrollment is such a relatively low-cost, extensive-margin measure for quality, however, that it makes sense that neither the heterogeneity nor non-existence of these previous studies shows up here. The significance of our coefficient is understandable given the many correlates of fertility and schooling amongst our regressors, whereas the small magnitude of our estimate may arise from a non-linearity in enrollment (e.g., if household size only begins to constrain at a high enough level), from economies of scale in enrollment costs, and from the aforementioned low quality threshold that is enrollment. This last factor also colors our interpretation of the relatively high probability of female enrollment (4.417 percent more than male enrollment): while it likely reflects the success of programs such as the Sarva Shiksha Abhiyan and the Kasturba Gandhi Balika Vidyalayas girls' schools (Press Information Bureau 2008), it also tells us nothing about the quality of education beyond enrollment.

While both Muslims and Christians are less likely to be enrolled than Hindus (whereas Buddhists and Zoroastrians are more likely), the sharp Muslim penalty of -12.604 percent echoes Borooah and Sabharwal (2021), who find that even upper-class Muslims enroll at lower rates. Even greater statistical significance can be found among Scheduled Tribe, Scheduled Caste, and Other Backward Class members, all of whom have negative coefficients,

[23] In 2002, the right to free and compulsory education for all children aged 6-14 was enshrined by Jain (2002) into the Constitution of India. One might expect this to explain our AME_{Age} result as the product of a steep drop in enrollment between the ages of 14 and 15 which violates the linearity assumption underlying the interpretation of AME values (Scholbeck et al. 2024). But the amendment neither enforced nor implemented, instead only mandating future legislation to do so; such legislation was not passed until 2009 Department of Higher Education (2010). Something which did exist in 2007-08, however, was the Sarva Shiksha Abhiyan program, which had been allocating funds for school construction, facilities management, teacher training, girls' education, and so on since 2001-02 (Press Information Bureau 2008). One would expect this to increase (as in make more positive) our $\text{AME}_{\text{Age}} = -2.834$ estimate, and it likely has increased our $\text{AME}_{\text{Female}} = 4.417$ estimate.

with Scheduled Tribes faring the worst at -10.217 percent. While we should be suspicious of causal interpretations, it is likely these coefficients reflect the way structural factors i.e., those which cannot be changed through individual actions alone, constrain the choice set over which households optimize over. For example, if we use chronic teacher absenteeism to reflect school quality, then we can use Kremer et al. (2005) to see that better infrastructure and proximity to a paved road correlate with increased school quality, as does more frequent school monitoring per both Kremer et al. and Muralidharan et al. (2017). If we assume that the physical and political infrastructure needed to do this is decreasing in the share of ST/SC/OBC members in the area, then this only decreases the opportunity cost of labor for children in these groups.

Looking deeper into the supply side of education, it seems that all variation which could potentially be explained by enrollment cost has, at best, been consumed by other variables, implying direct costs matter less than social barriers and other correlates of supply-side factors. While one would expect the causal effect of an exogenous shock in scholarships/stipends, stationery, or textbooks to match the sign of their positive coefficient estimates, only textbooks show up as statistically significant. Instead, our other significant result is the -8.856 percent change in enrollment associated with a marginal change in the share of students within one's district who attend a school which charges no tuition fees (i.e., where 'Educ. freely available'). National Sample Survey Office (2011) documentation indicates that such schools include most if not all government schools, as well as private schools in some states up to a certain level of education. Building off observations in Section 1 that government schools tend to have worse facilities, chronic teacher absenteeism, and so on, this coefficient seems to imply that when the modal inside option is poor-quality schooling, the marginal child is more likely to have taken the outside option not be enrolled. If this interpretation is correct, then comparing the variables' s-values[24] using Mansournia et al. (2022) reveals just how much evidence we find for government schools being a negatively-priced deterrent (a la Muralidharan 2019): while both 'Textbook(s) received' and 'Educ. free available' have similar effect sizes and standard errors, the former has an s-value of 12.6, meaning that the data provided 12.6 bits of information against the null hypothesis (a coefficient of zero), whereas the 'Educ. free available' result communicated a far higher 27.5 bits against its null.

Finally we have the most economically and statistically significant terms: the father's (or father proxy's) education level. We can interpret this in multiple ways: as a proxy for financial ability (via the relationship between a father's education and the household's permanent income), for heritable cognitive ability, for a child's valuation of education (via intergenerational transmission of tastes and values), and even for access to educational and labor market information (via social capital). It's possible that these variables, insofar as they correlate with fertility and schooling choices, absorbed some of the variation which otherwise would've been explained by, say, household size. But the size and significance of these coefficients may imply something further: that

[24] Given a p-value, we can define the s-value as $s = -\log_2(p)$ (Mansournia et al. 2022).

a household's financial/cognitive ability, tastes/preferences, and social capital/information networks affect both household optimization and the choice set available for them to optimize over.

4 The Effect of EMIE: 2SLS Estimation and Main Results

4.1 Model and Variables

Inspired by the measures of public school exposure in Chakraborty and Bakshi (2016), we define district EMI exposure (EMIE) as the survey-weighted share of children ages 5–19 who are both enrolled and observed in English-medium instruction:

$$\text{EMIE}_d = 100 \times \frac{\text{Num. children ages 5–19 enrolled in EMI}}{\text{Num. children ages 5–19}} \text{ in district } d.$$

This all-child exposure combines the extensive margin of school enrollment with the medium-of-instruction choice among enrolled children. We are interested in how variation in this exposure affects district real consumption growth from 2007-08 to 2017-18. Our preferred outcome is the log difference in person-weighted real monthly consumption. Nominal household expenditures are first deflated using state-sector temporal indices and Tendulkar spatial relatives, then aggregated to districts with household survey weights and household size. We use LingDistance_d as an instrument for EMIE_d (see Section 4.2). The public 2SLS specification is

$$\begin{aligned} \text{EMIE}_d &= \gamma_0 + \gamma_1 \,\text{LingDistance}_d + X_d'\gamma + \alpha_s + v_d, \\ \Delta \log C_d^{real} &= \beta_0 + \beta_1 \,\widehat{\text{EMIE}}_d + X_d'\beta + \alpha_s + u_d, \end{aligned} \tag{2}$$

where X_d contains predetermined Census 2001 controls and α_s denotes Census-2001 state fixed effects. Summary statistics for all variables in this model, including the controls k in the vector X_{kd}, are provided in Table 4. Maps of these variables, created using 2001 boundaries, are presented in Figure 2 and Figure 3.

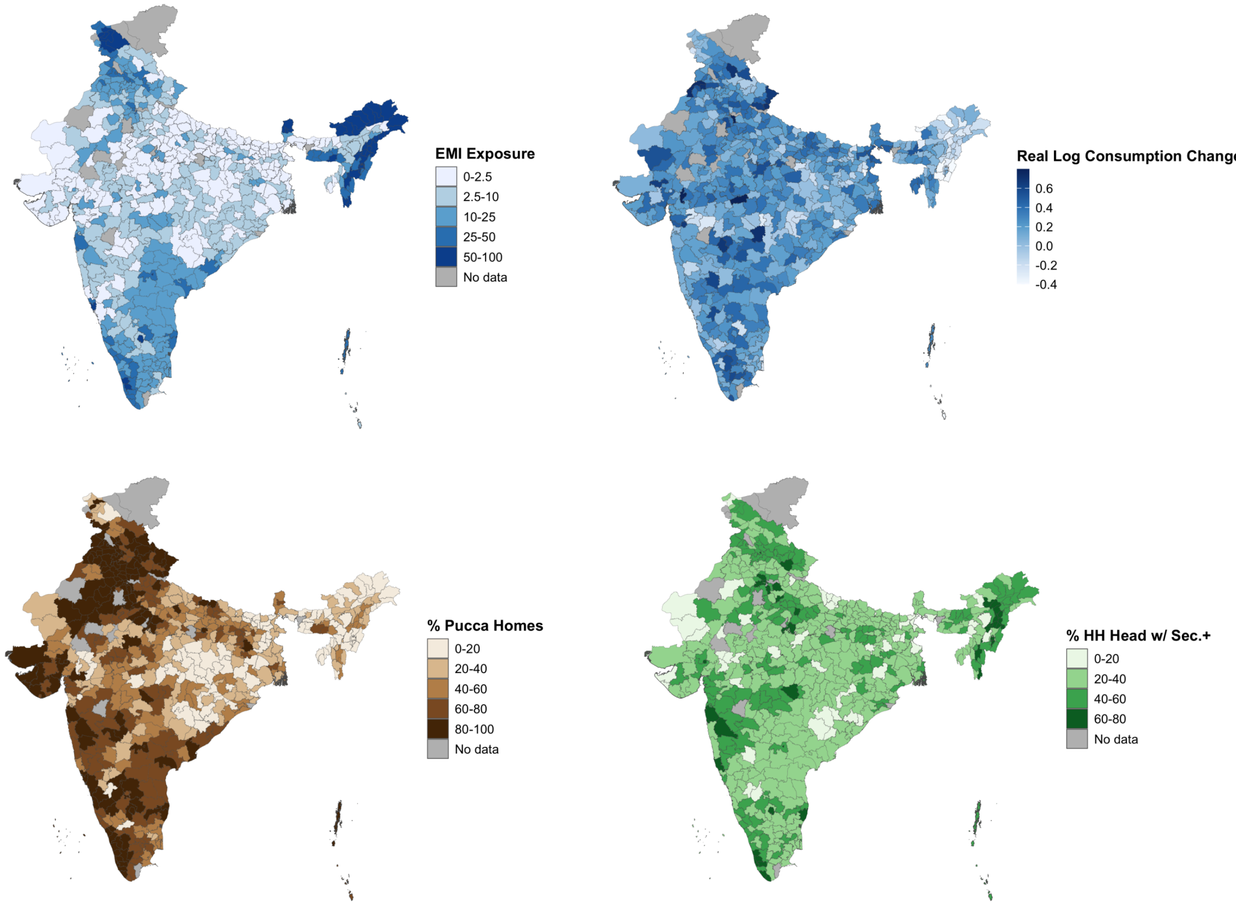


Figure 2: (Clockwise from top left) EMI exposure, consumption growth, pucca (permanent) housing, and household heads with secondary education or more. Data from the 64th round of the NSS 2007-08, "Participation and Expenditure in Education" and "Household Consumer Expenditure."

Table 4: Summary Statistics for 2SLS Model

Variable	Description	Min	1Q	Med	3Q	Max	Mean	SD	N
Treatment and instrument:									
Linguistic distance	Population-weighted mean linguistic distance among mapped speakers with positive distance from Hindi	1.00	1.97	3.02	4.33	5.00	3.05	1.36	573
EMI exposure	Share of children ages 5-19 enrolled in English-medium instruction	0.00	1.42	6.23	17.91	95.98	14.90	20.35	573
Consumption outcomes:									
Real consumption, 2007-08	Person-weighted monthly consumption in common prices	439.46	826.48	986.00	1179.91	2945.06	1035.78	295.72	573
Real consumption, 2017-18	Person-weighted monthly consumption in common prices	561.61	996.59	1174.29	1418.34	4170.07	1261.98	398.55	573
Real log consumption change	Log real consumption in 2017-18 minus log real consumption in 2007-08	-0.86	0.07	0.20	0.34	0.98	0.19	0.22	573
Census 2001 controls:									
Log population	Natural log of Census 2001 district population	10.35	13.61	14.20	14.68	16.08	14.00	1.02	573
Urban population share	Urban population as a percentage of total population	0.00	10.03	18.15	29.73	100.00	23.27	19.11	573
Secondary-plus share, age 7+	Population age 7 and above with matric or higher attainment	4.40	11.39	15.62	21.08	43.67	17.09	7.63	573

Table 4: Summary Statistics for 2SLS Model *(continued)*

Variable	Description	Min	1Q	Med	3Q	Max	Mean	SD	N
Scheduled Caste share	Scheduled Caste population as a percentage of total population	0.00	8.13	15.64	19.98	50.11	14.71	8.64	573
Scheduled Tribe share	Scheduled Tribe population as a percentage of total population	0.00	0.15	3.49	18.27	98.09	16.16	25.85	573
Muslim share	Muslim population as a percentage of total population	0.09	2.59	7.21	13.51	98.49	11.40	14.81	573
Agricultural worker share	Cultivators and agricultural labourers as a percentage of workers	0.00	50.56	66.32	75.37	90.24	60.53	20.40	573
Dependency ratio	Population age 0-14 and 65+ as a percentage of population age 15-64	35.46	58.79	69.09	80.75	100.75	69.39	13.29	573
Electricity access share	Households using electricity for lighting as a percentage of households	3.09	28.36	60.14	78.40	99.71	54.44	28.01	573
Log population density	Natural log of persons per square kilometre	0.88	5.23	5.76	6.41	10.82	5.77	1.16	573

Note: Min. = minimum; 1Q = first quartile; Med. = median; 3Q = third quartile; Max. = maximum; Mean = arithmetic mean; SD = standard deviation; N = number of observations.

4.2 Instrumental Variable

Our instrument for EMIE_d is LingDistance_d, the speaker-weighted mean Shastry distance from Hindi among mapped mother tongues with positive distance in district d in 2001. The mechanism behind this may seem odd: why choose linguistic distance from Hindi as opposed to, say, English? India's large linguistic diversity[25] predisposes it to have large communication costs, large coordination costs, and thus hindered economic development within its borders (Nakagawa and Sugasawa 2021). To decrease such frictions in the market and in public goods provisioning (Liu and Pizzi 2016), both Hindi and English have been thrust into the roles of competing domestic lingua franca (Clingingsmith 2014), meaning Hindi- and English-medium schools can be found across the country (Shastry 2012, 294).

As the 'distance' between one's mother tongue and Hindi increases, we expect the opportunity cost of learning English over Hindi to decrease. Because of this, we also expect individuals with a mother tongue further from Hindi to be more likely to enroll in EMI. "That distance from Hindi induces people to learn English" and even "schools to teach English" is in fact precisely what Shastry (2012, 289) claims. If we can then show that local linguistic distance induces exogenous variation in local EMI, we can wield it as an instrumental variable (IV) to isolate EMIE_d's effect on consumption growth. Shastry helps us once more here, evidencing both the relevance (pp. 299-301) and exclusion restriction (pp. 302-305) of LingDistance_d's effect on EMIE_d.

We are currently unable to replicate her justification of the exclusion restriction, however. Her argument centers on a map depicting the geographical balance of her residual variation, made possible despite the distinct regional divide in Figure 3 thanks to state fixed effects. In our case, the state-FE specification has condition number $\kappa = 24.92142$ even though the production pseudo-panel uses unique Census-2001 target districts and passes the lineage validation gates. The remaining instability is therefore an identification problem rather than an artifact of duplicated panel rows: linguistic distance is strongly structured across states, leaving little conditional variation once state fixed effects and predetermined district controls are included.

[25] Perhaps the largest in the world, per one metric from Gurevich et al. (2025).

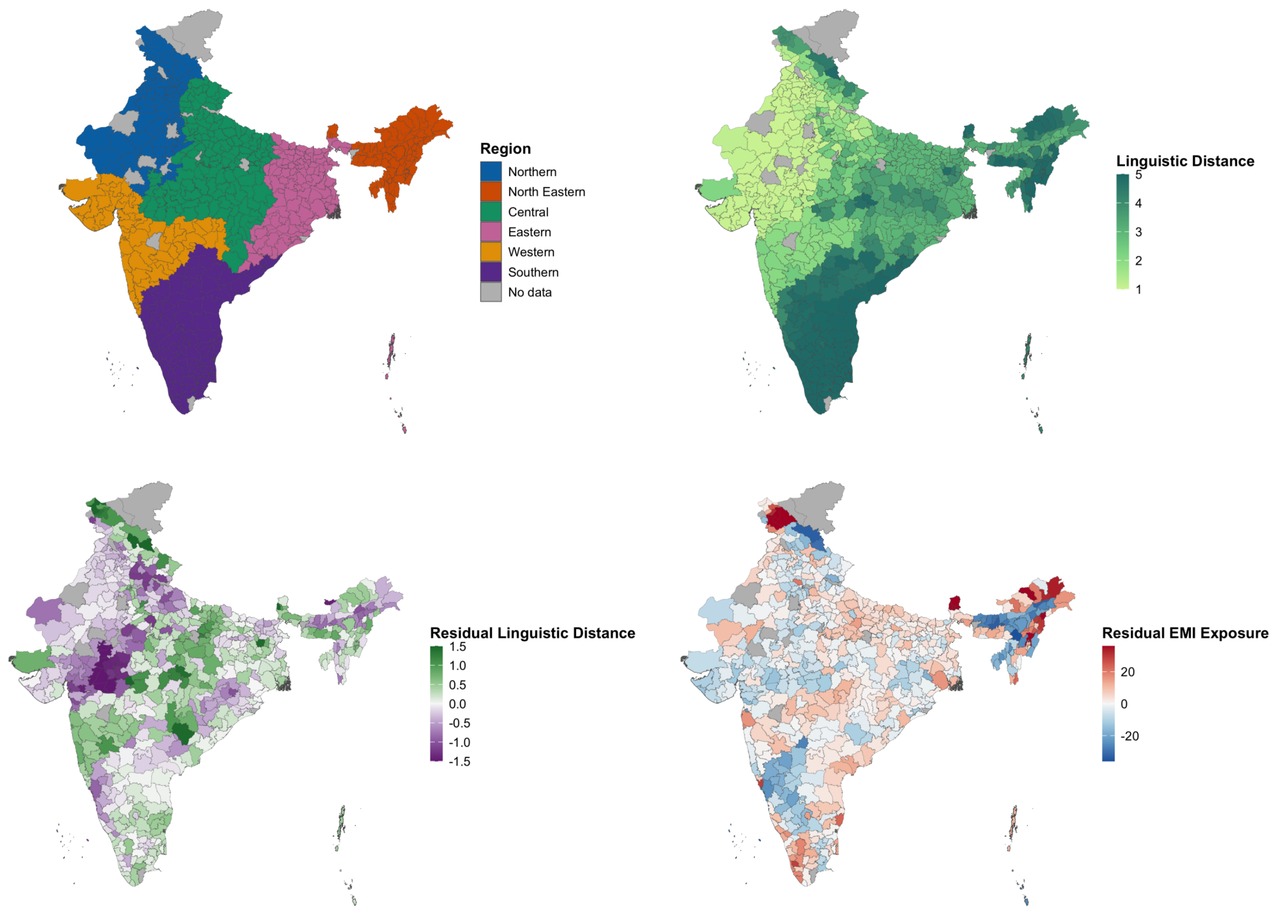


Figure 3: From left to right: regions of India and linguistic distance from Hindi. District-level data, from the 2001 Census of India.

Part of this absorption is structural: the States Reorganization Act of 1956 aligned many state lines with linguistic lines, a precedent later states have often followed (Sarma 2025). This is why the state-FE first stage is an important stress test rather than a nuisance to be removed. Future work should seek excluded variation that survives within-state conditioning, potentially through time-varying supply or demand shocks and weak-IV-robust estimators rather than dimensionality-reduction methods that change the identifying variation.

In the meantime, we rely on heuristic justifications of our instrument. Its relevance is plausible: districts with higher linguistic distance from Hindi should have higher EMI demand due to the relatively lower opportunity cost of learning English. They should also have relatively lower cross-language coordination costs for setting up EMI as opposed to Hindi-medium instruction, allowing EMI supply to rise more. Shastry justifies this at the state level; future work must replicate this at the district level.

The exclusion restriction, meanwhile, requires us to assume that after conditioning on our controls, a district's linguistic distance is a) predetermined and b) plausibly orthogonal to all later shocks and trends in consumption except via education. We could demonstrate (b) by replicating Shastry's visualization of geographically balanced residuals, though this will require state/region fixed effects (and possibly spatial correlation robust SEs; see Section A.2.2). More work will be needed to replicate her justification of (a): not only does she show that district-level linguistic distance in both 1991 and 1961 are strongly correlated, she also shows that they both increase the within-state share of a language's native speakers who learn English in 2001, as well as the in-state level of EMI supply in 2001. That linguistic distance remains consistent over decades implies little inter-district migration, in line with most literature in Section A.2.2.[26] Our exclusion restriction may still fail, however, if districts further from Hindi benefit differently from pre-existing human capital, access to non-Hindi media, trade relations, NGO/government funding, and so on compared to districts closer to Hindi. Further work will be needed to address this.

LingDistance_d uses Shastry's simplest and most frequently used measure: a 0-5 integer scale reflecting the "degrees" of separation between one's mother tongue and Hindi, developed with Harvard linguist Jay Jasanoff.[27] We assign those published distance categories through the tracked Census-2001 language concordance, then take the speaker-weighted mean among mapped languages with positive distance in each district. Coverage and alternative constructions, including the older top-three-language mean and the five nonzero distance shares, are reported in extended diagnostics rather than substituted into the public specification.

4.3 Results

The results of our first-stage regression, of EMI exposure on linguistic distance, are provided in Table 5a. The results of the second-stage regression, where real log consumption growth is regressed onto the fitted values of EMI exposure, are provided in Table 6a. All standard errors in both stages are clustered at the state level.

4.4 Discussion

The preferred specification uses person-weighted real log consumption growth as the outcome, the predetermined Census 2001 control set, and Census-2001 state fixed effects. The excluded-instrument partial F is 0.78 with $p = 0.38$. The first-stage coefficient on linguistic distance is 0.85. Conditional relevance is therefore extremely weak.

The second-stage EMI-exposure estimate is 0.04 with $p = 0.454$. Because the instrument contributes almost no conditional first-stage variation, this estimate should not be interpreted as a credible causal effect. The

[26] Though perhaps not in line with Economic Division (2017), whose gravity model for migration purportedly shows that while political barriers constrain migration, linguistic ones do not: "not having a common language does not impede the flow of migrants" (p. 275). Their 'common language' proxy, however, only indicates if a state is primarily Hindi-speaking or not. If Hindi truly does compete with English as national lingua franca, as Clingingsmith (2014) claims, then insignificance is expected.

[27] Shastry (2012) finds similar results across different measures, including the share of speakers with mother tongues three or more degrees away. One such measure relies on currently inaccessible data on the percent of cognates shared with Hindi. Anderson et al. (2025) may enable us to construct a cognate-based robustness measure.

Table 5

First-Stage Regression: EMI Exposure on Linguistic Distance

	EMI Exposure
	(1)
Linguistic distance	0.847
	(0.962)
Log population	0.686
	(0.718)
Urban population share	0.117*
	(0.052)
Secondary-plus share, age 7+	0.563***
	(0.166)
Scheduled Caste share	0.038
	(0.067)
Scheduled Tribe share	0.044
	(0.031)
Muslim share	0.056
	(0.042)
Agricultural worker share	-0.047
	(0.054)
Dependency ratio	-0.049
	(0.067)
Electricity access share	-0.047
	(0.043)
Log population density	1.012
	(0.800)
Intercept	29.012**
	(9.330)
Observations	573
R^2	0.894
Adjusted R^2	0.885
Residual Std. Error	6.890
Model's F-Statistic	99.20
Instrument's F-Statistic	0.78

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Standard errors clustered by state in parentheses.

Table 6

Second-Stage Regression: Real Log Consumption Growth on EMI Exposure (Fitted)

	Real Log Consumption Growth
	(1)
EMI exposure (fitted)	0.036
	(0.049)
Log population	-0.011
	(0.040)
Urban population share	-0.000
	(0.006)
Secondary-plus share, age 7+	-0.019
	(0.028)
Scheduled Caste share	-0.003
	(0.004)
Scheduled Tribe share	-0.003
	(0.003)
Muslim share	-0.001
	(0.003)
Agricultural worker share	0.003
	(0.003)
Dependency ratio	0.008*
	(0.004)
Electricity access share	0.001
	(0.003)
Log population density	-0.063
	(0.069)
Intercept	-1.727
	(1.674)
Observations	573
R^2	-0.961
Adjusted R^2	-1.128
Residual Std. Error	0.327
F-Statistic	

* p < 0.05, ** p < 0.01, *** p < 0.001

Standard errors clustered by state in parentheses.

appropriate conclusion is not that EMI has no effect, but that this instrument cannot distinguish such an effect under the preferred control and fixed-effect specification.

The revised design improves measurement and temporal ordering. Consumption is adjusted for state, rural/urban, and survey-month price differences; district means represent people rather than households; controls are measured before treatment; and district histories are harmonized to a Census-2001 geography. Those improvements expose rather than solve the identification problem: most of the raw linguistic-distance relationship with EMI is between states or is shared with predetermined district characteristics.

Future identification work should therefore focus on alternative excluded variation, transparent reduced-form evidence, leave-one-state-out first stages, and weak-IV-robust confidence sets. Nominal log change, real ANCOVA, alternative lineage panels, and the legacy control set remain diagnostic comparisons rather than substitutes for instrument relevance.

Migration and spatial spillovers remain important for interpretation. EMI may raise the welfare of students who later leave their childhood districts, while the district-level outcome records local equilibrium changes rather than individual returns. Remittances, firm location, peer effects, and school-supply responses can therefore move the district estimate away from the individual effect of EMI.

The Census controls reduce concern that the instrument is merely proxying for baseline population, urbanization, education, caste and religious composition, agricultural employment, age structure, electricity access, or density. They do not establish the exclusion restriction, and their joint inclusion with state fixed effects leaves the first stage too weak for conventional 2SLS inference. Spatial diagnostics and lineage-panel sensitivity analyses are reported separately, but neither can restore identifying variation that is absent from the first stage.

A Appendix

A.1 Alternative Response and Further Control Variables

The preferred response variable is the change in log person-weighted real consumption between 2007-08 and 2017-18. Household expenditures are assigned state-sector temporal price indices by NSS sub-round and are spatially adjusted using Tendulkar state-sector poverty-line relatives. Nominal log change and a real-level ANCOVA specification are retained as fixed-sample comparisons. These adjustments address measured price variation, but they cannot remove unobserved district-specific economic shocks or differences in the consumption instruments used across survey waves.

In the meantime, other response variable such as $\%\Delta\text{Wages}_d$, the percent change in district d's average household wages from 2007-08 to 2017-18, sourced from the "Employment, Unemployment, and Migration" survey of 2007-

08, another part of the 64th Round of the NSS (National Sample Survey Office 2010), as well as from the Periodic Labor Force Survey (PLFS) of 2017-18 (National Statistical Office 2019), may be informative. On one hand, the stickiness of wages may control for potential volatility in $\%\Delta\text{Consumption}_d$. On the other hand, wages may not be affected by the lifelong consumption smoothing underlying $\%\Delta\text{Consumption}_d$—in other words, wages be less responsive to unobserved shocks while still being *more* responsive to any effect of EMIE_d which may manifest in the decade-long window we study. None of this diminishes the fact that informal, self-employed, and otherwise non-wage laborers are very common in India. Individuals vary in the degree to which wages reflect their true income, and this variation may not be exogenous to our model. Numerous papers studying similar contexts in India have still used wages as their primary response variable (Shastry 2012; Azam et al. 2013; Madheswaran and Singhari 2016; Refeque and Azad 2022). $\%\Delta\text{Employment}_d$, the percent change in district d's employment, sourced from National Sample Survey Office (2010) and National Statistical Office (2019), may provide additional insights as well.

$\Delta\text{Gini}_d^{\text{Consumption}}$ and $\Delta\text{Gini}_d^{\text{Wages}}$, the change in the Gini coefficient of district d's consumption and wages respectively from 2007-08 to 2017-18, would share the same sources as their non-Gini counterparts above. The models of Equation Equation 2 were actually estimated using $\Delta\text{Gini}_d^{\text{Consumption}}$ as the response variable; this returned a near-zero R^2 and adjusted R^2, meaning the model did no better than the sample mean. Within-district distributional effects of EMIE, if they exist, almost certainly need more than 10 years to manifest.

$\%\Delta\text{HDI}_d$, the percent change in district d's Human Development Index (HDI), sourced from the National Family Health Surveys of 2005-06 (NFHS-III) and 2019-21 (NFHS-V), would provide a more holistic, shock-resistant measure of development, even if the developmental effects of EMI may need more than 15 years to reveal themselves and even if better alternatives (e.g., the HDI_a of Chaurasia 2023) exist. But NFHS data has unfortunately been unavailable since the USAID funding cuts of 2025.

To further control for omitted variables and to further reveal the mechanisms behind EMIE's effect on consumption, future work may include TechFirms_d, a measure of the number of IT firms from the 2005 Economic Census (Central Statistics Office 2008);[28] infrastructure measures beyond the proportion of pucca (i.e., permanent) housing; and proxies for corruption, chronic teacher absenteeism, etc., all of which are assumed to be similar across districts. Shastry (2012) incorporates corruption by including a "Hindi Belt" indicator for states such as Bihar, Delhi, Uttar Pradesh, and Rajasthan, all majority Hindi-speaking states known for their "high levels of corruption and government inefficiency" (p. 299). Note that a desire for EMI *and* a desire for better teacher quality among poorer households is what drove a 30% increase in the number of private schools from 2012 to 2020, most of them low-cost private schools (Gooptu and Mukherjee 2023, 2). This did not seem to change the fact that private schools attract richer households than government schools on average (Muralidharan and Sundararaman 2015),

[28] State fixed effects should absorb many predictors of this e.g., labor regulation (Besley and Burgess 2004).

but it does imply a relationship between cost, private vs. government schools, and teacher quality which may inform future proxies of teacher quality. Adding in a control for migration flows will be key as well, to ensure that the effect of EMIE does not dissipate above a certain threshold due to all the out-migration to urban hubs it has facilitated. See Section A.2.2 for more on migration and spatial effects.

Interaction terms such as $\text{EMIE}_d \times \text{Urban}_d$ and $\text{EMIE}_d \times \text{TechFirms}_d$ may also be key in identifying the mechanisms by which EMIE can affect a district's growth. Even if households have identical underlying preferences for how to educate their children, the menu of choices available to them (that is, the feasible set over which they optimize) may still be affected by factors far larger than their own education-related decision-making—whether that be by the presence of IT firms nearby, or by the parents' familiarity with and ability to learn more about the Indian education system. To better understand the meaning of the highly significant coefficient on urbanization in Table 6a, the urbanization and EMI interaction may be complemented with an urbanization and parental/household head education interaction. The low Muslim/ST/OBC estimates, too, can be studied further by interacting them baseline consumption shares, urbanization, and EMIE itself. And finally, to understand the small coefficients and relative insignificance of the household head's education estimates, we could interact these education levels with urbanization and baseline consumption, or we could even try replacing our baselines with changes, as with a first-difference estimator.

Notice how all actual and proposed regressors in this model reflect baseline exposures. Replacing these with changes and time-varying instruments instead may allow us to use a Bartik-style methodology (e.g., Goldsmith-Pinkham et al. 2020). Linguistic distance, for example, could be turned into a time-varying instrument if interacted by a nation or state-wide shock in EMI supply or demand between 2007-08 and 2017-18. We have yet to identify shocks which have both a plausible effect on EMIE (e.g., changing curricular standards or waves of English training for teachers) and which vary across time alone, not district.

A.2 District Matching and Spatial Autocorrelation

A.2.1 District Matching Method

There are three reasons that district labels vary between 2001 to 2007-08 to 2017-18 in the data: genuine political changes, variation in anglicization, and typos. Fuzzy joins (described below) were used to correct for the latter two.

Political changes, however, were not so easily addressed. The vast majority of such changes were one of five kinds: name changes, clean partitions (one district is partitioned into multiple), clean mergers (multiple districts merge into one), carve-outs (*pieces* of multiple districts combine into one), and boundary shifts (India State Stories 2025). Without knowing precisely where in its district each household was at the time of sampling, it is not feasible to match districts across carve-outs and boundary shifts. Our district matching method is thus reliant on the assumption that all district changes were either name changes, clean partitions, or clean mergers.

This seems to be precisely the same assumption made by India State Stories (2025) and Jaacks et al. (2020), whose data, despite containing some factual errors and typos, was the basis of our district matching method. To justify this assumption we turn to Kumar and Somanathan (2016), who up until 2001 are able to track the *proportion* of each district's population that was allocated into a new district as a result of district changes. They label these changes as one of two types: clean partitions and non-partitions (which would include name changes, clean mergers, carve-outs, and border shifts). By extracting their results using Tabula (Aristarán et al. 2018), we know that 93.4% of non-partitions were either name changes or transfers of uninhabited land (Kumar and Somanathan 2016). These district changes are plotted in Figure 4.

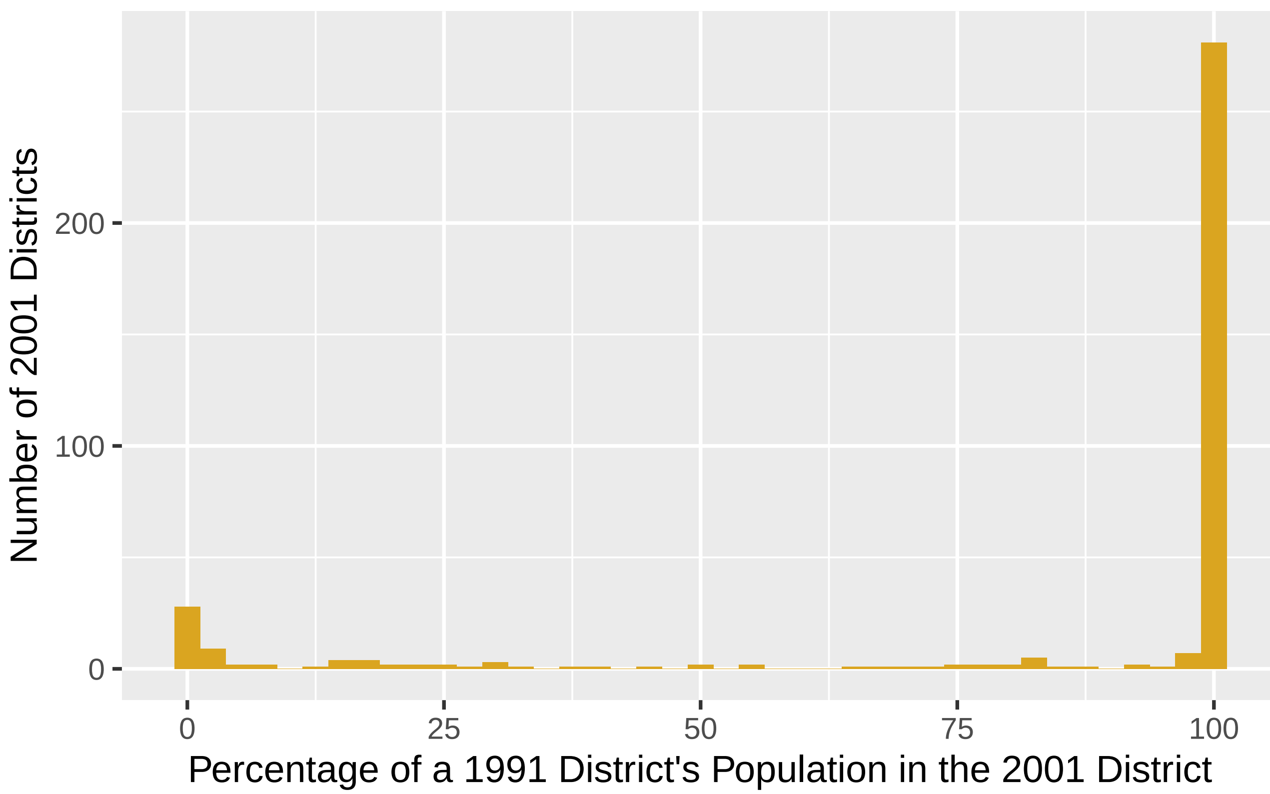


Figure 4: Number of 2001 districts which absorbed a percentage of a 1991 district's population via name change, clean merger, carve-out, or border shift. Data from Kumar & Somanathan (2016).

We thus assume no district changes were carve-outs or border shifts. To match districts, we ran a hierarchical, cascading sequence of iterated fuzzy joins. Data from 2001, 2007-08, and 2017-18 is first linked to the closest year in the district trackers of India State Stories (2025) and Jaacks et al. (2020). Districts are then matched if they meet a certain string/phonetic distance threshold. Matches were made in the following order:

1. `soundex` = 0, for phonetically equivalent variations in anglicization (e.g., "Baleswar" and "Balasore").
2. `qgram` distance = 0, for rearrangements of bigrams (e.g., "East Godavari" and "Godavari East").
3. Jaro-Winkler distance ≤ 0.15, for respellings and vowel swaps with 0.85 degrees of similarity.
4. Full Damerau-Levenshtein distance ≤ 2, for ≤ 2 insertions, deletions, substitutions, and transpositions.
5. Optimal String Alignment distance ≤ 1, for a single typo.

Any district from the 2001, 2007-08, or 2017-18 data which remains unmatched after this process is then matched with the next closest year in the district tracker. This process repeats until all years in the tracker had been exhausted. 80 still-unmatched districts were then manually matched.

The active harmonization now uses Census-2001 districts as the common unit. Counts are summed, means and shares are recomputed from pooled numerators and denominators, and Gini coefficients are reconstructed from pooled household records where support permits. Conservative and broader reviewed lineage panels are retained as sensitivity analyses. State fixed effects are now estimable, but the excluded instrument has almost no conditional relevance once those fixed effects and the Census controls are included.

A.2.2 Spatial Autocorrelation, Spatial Spillovers, and Migration

The maps in Figure 2 and Figure 3 now use Census-2001 district geometry for the harmonized pseudo-panel. Districts absent from the analysis maps lack complete two-wave analysis inputs or belong to the small set of explicitly non-identified reorganizations; treatment values are no longer propagated across 2019-20 successor districts.

The presence of true spatial autocorrelation in this environment is plausible. The spatial spillovers of development and the agglomeration of tech firms are key forms of autocorrelation which render the assumption of identical and independent distributions false, and thus many of the results of OLS meaningless. But special attention must be paid to the role of migration here. Note that, if part of EMI's effect is in enabling children to migrate across the country (e.g., to districts which have more tech firms), then the true structural parameter EMIE_d in our model would only reflect the EMI-induced benefits which they send back e.g., as remittances. In other words, even in a world of perfect shapefiles, our results could show that EMIE is a substandard tool for local area's development even if it is actually a great tool for a local population's development. How, then, could the effect of EMI ever be identified without being dissipated away by between-district migration?

Chakraborty and Bakshi (2016) find a 4% decadal rate of inter-district migration in their states of interest (West Bengal, Haryana, and Punjab) up until 2001, a migration rate[29] so low that they assume migration away almost

[29]We define an area's 'migration rate' over a period as the share of this area's end-of-period population who report having migrated

entirely, effectively equating the district where one's education occurred with the district most affected by their education.[30] Topalova (2005) find decadal inter-district migration rates to be around 3.2% and 13.1% of rural and urban areas respectively, or around 0.5% and 4.0% when only considering migration related to employment, trends which were "remarkably constant" across her years of study (1983, 1987, and 1999), including "no visible increase after the economic reforms of 1991" (p. 24), which preceded to the IT growth and service job offshoring described in Section 1; National Statistical Office (2021) imply not much has changed since either, albeit using 2020-21 data. It seems to be a structural, secular trend in India that migration only occurs over short distances.[31]

Why could this be the case? Topalova (2005) identifies marriage as the largest factor behind migration writ large, but this effect is driven entirely by women:[32] using 2020-21 data, the National Statistical Office (2021) demonstrate that male migration is instead driven almost entirely by economic factors. For example, Kochar (1999) finds that after negative farm income shocks, any consumption loss households would face is offset by men taking on more off-farm work, implying that local labor markets in rural India are sufficiently diversified, local social capital is sufficiently anchoring, and other markets are sufficiently segmented off for households to consumption smooth using at most short-distance, temporary migration. "Households use migration as an ex post income-smoothing mechanism," states Morten (2019, 3), whose structural model on the interplay of risk sharing and migration (particularly rural-urban migration) corroborates the role of risk in limiting such migration just as Munshi and Rosenzweig (2016) corroborate the roles of risk and social capital; the latter authors find caste-based networks of rural insurance significantly constrain rural males' ability to migrate, and structural estimation indicates a large effect on migration from marginal improvements in formal insurance, particularly on rural-urban migration. Administrative differences may also play a factor, though Kone et al. (2018) imply that such differences matter far more across states' borders than districts'; by using 2001 census data, in fact, they find rates of long-distance inter-*state* migration to be far lower than those of Brazil and China, which they posit is the product of non-portable state-level welfare schemes plus states' preference to hire their own residents in the public sector.

Newer findings, however, repudiate many of these low migration rates wholesale. Lucas (2021a) finds that, using

over said period. The 'decadal inter-district migration rate for rural areas in 2001', for example, would refer to the "percent of people living in rural areas [who in 2001] reported changing [their district] within the past 10 years" (Topalova 2005, 25). Social scientists only look at migrations reported within the past one, five, ten, etc. years to both capture shorter-term fluctuations in migration trends and to avoid the cumulative nature of metrics built on all lifetime migrants (Mahapatro 2020).

[30] More precisely, they "assume that district of current residence (or of employment) of an individual is approximately the same as the schooling district" (Chakraborty and Bakshi 2016, 6).

[31] And perhaps only for short times as well, though this may only be the product of data interpretation errors. Keshri and Bhagat (2013), for example, use 2007-08 data from the National Sample Survey Office (2010) to find a temporary (1-6 months) labor migration rate of 26% in rural areas, 13 times larger than the permanent labor migration rate. And yet, despite using the same 2007-08 data, Lucas (2021b) finds this same rate to be 2.6%. Barring the possibility of a misplaced decimal point, the sole difference seems to be Lucas's observation that the NSS Office's definition of 'temporary' migration only considered if a migration was *intended* to be temporary, not if it actually was; he adjusts his measures accordingly, whereas Keshri and Bhagat, it seems, do not. Imbert and Papp (2020) corroborate Lucas's number.

[32] Though a large majority of women list marriage as their primary reason for migrating, over the past few decades economic factors have gained greater and greater explanatory power (Mahapatro 2020), and women who list marriage will increasingly begin working right after migrating anyways (Rajan et al. 2020). The relationship between social capital, information networks, and economically-motivated female migration is further explored by Neetha (2004).

2007-08 data from the National Sample Survey Office (2010), the 4.7% net flow of migrants from rural to urban areas in 2003-08 was among the highest of all countries studied.[33] More novel methods, however, were used by the Economic Division (2017). Their cohort-based migration metric[34] finds the inter-district migrant share alone to be 6.6% when applied to 2001 and 2011 census data. Railway passenger data is then used to proxy for labor-related migration flows which are found to extend well beyond even state boundaries, almost entirely between large urban hubs. The final finding relevant to us, shared by both the Economic Division (2017) and the more standard analysis of Mahapatro (2020), is that the stagnant migration rates of the late 20th century, as observed by Topalova (2005), have increased since 2001, perhaps the product of economic growth which has raised the benefits of migration increasingly above and beyond its costs (Economic Division 2017).

The significance of this new literature for us is two-fold: one, it means the mechanism behind part of Shastry (2012)'s exclusion restriction demonstration in Section 4.2 may not apply as much to our period of interest; and two, it means that EMIE's effect dissipating away with migration is a genuine possibility. If we tentatively assume that regressors such as urbanization adequately capture the effect of longer-distance migration (which accords with the Economic Division 2017's finding that it is "rich metropolises [attracting] large inward flows of labour," p. 277) then the most relevant kind of dissipation for us will be over neighboring district lines (which also accords with Economic Division 2017, 276).[35]

To test for spatial autocorrelation, we follow p. 323 of Anselin (2001) and construct Moran's I statistics on the harmonized Census-2001 district geometry. Using rook-contiguity, row-standardized weights for the active 2SLS sample, the p-value for spatial autocorrelation in the residuals of our consumption model is 0.427 and for consumption growth is 4.62e-31.[36] These diagnostics now use the same geographic units as the pseudo-panel rather than later successor-district geometry.

If spatial autocorrelation is ultimately found to be significant, then the methods of Murgante and Borruso (2012), Pebesma and Bivand (2025), Xu and Wooldridge (2022), and Zhang et al. (2024) would be used to develop a spatial Durbin-type model (a spatial error model, a spatial lag model, a spatial Durbin model, a general nested model, etc., depending on the results of LM tests on the common factor specifications which distinguish these kinds of models), a class of model which incorporates spatial lags of spatially autocorrelated variables as further regressors or in the error. Conley standard errors and/or Kelejian and Prucha (2007) could also be used to create heteroskedasticity and autocorrelation consistent SEs.

[33] Lucas (2021b) also uses NSS data to find far, far smaller rates of short-term migration, with 2.6% of rural adults migrating for 1-6 months versus the 50% estimates cited by, say, Morten (2019).

[34] Which defines net migration as "the percentage change in population between the 10-19 year-old cohort in an initial census period and the 20-29 year-old cohort in the same area a decade later, after correcting for mortality effects" (Economic Division 2017, 267).

[35] Reminiscent of Kone et al. (2018), the Economic Division (2017) finds that within-state flows of labor migration are much higher than between-state flows. State fixed effects will be essential for further analysis.

[36] Queen-contiguity comparisons and connectivity diagnostics are retained in the extended spatial diagnostics.

B Technical Note for Replication

Those replicating the code for this paper are encouraged to skim the code chunk titled `Define functions to read in short 8.3 filenames`. There the functions `read_sav_short()`, `read_csv_short()`, and `read_excel_short()` are defined; if the replicator runs into issues while reading in files, troubleshooting methods are explained there.

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